10th grade
Learning evidence 2.1: Analysis of Decisions (Practice)

Evaluated criteria

This exam evaluates the following criteria. Each criterion is evaluated across the items below. A student passes a criterion if it is correctly applied in the solution.

- C2 Interprets decision alternatives, events, consequences, and states of nature.
- C3 Builds a payoff table from a description of the problem.
- C4 Explains the maximax criterion for decision making without probabilities.
- C5 Summarizes the maximin criterion for decision making without probabilities.
- C1 Describes how a problem can be formulated for optimal decision making, computes the expected value, and interprets the result.

Problem 1: A community recreation center must choose a membership model for the next year: Model A (standard monthly pass), Model B (premium pass with classes), or Model C (flex pass). Demand for memberships can be high, medium, or low, and staffing costs can be favorable or unfavorable. The demand probabilities are $0{,}30$ (high), $0{,}45$ (medium), and $0{,}25$ (low). Staffing costs are independent of demand, with probabilities $0{,}55$ (favorable) and $0{,}45$ (unfavorable). Expected membership counts depend on demand and model: under high demand the center expects 640 members with Model A, 500 with Model B, and 720 with Model C; under medium demand it expects 480 with Model A, 380 with Model B, and 540 with Model C; under low demand it expects 320 with Model A, 250 with Model B, and 380 with Model C. Membership fees are 36 for Model A, 50 for Model B, and 32 for Model C per member per year. Variable staffing cost per member is 19 when costs are favorable and 27 when costs are unfavorable. Fixed annual costs are $6{,}800$ for Model A, $9{,}200$ for Model B, and $5{,}400$ for Model C. All monetary amounts in this problem can be treated as dollars. Use the combined demand and staffing outcomes as states of nature and the stated probabilities to compute expected profits.

Problem 2: A school club must choose one of three snack plans for a two-week fundraiser: Plan A (fresh-made items), Plan B (mixed menu), or Plan C (budget staples). Demand for snacks can be high, medium, or low. Based on past fundraisers, the probability of high demand is $0{,}30$, medium demand is $0{,}50$, and low demand is $0{,}20$. Profits are measured in hundreds of dollars. If demand is high, Plan A yields 88, Plan B yields 74, and Plan C yields 60. If demand is medium, Plan A yields 54, Plan B yields 66, and Plan C yields 50. If demand is low, Plan A yields -6, Plan B yields 28, and Plan C yields 36. The stated probabilities must be used for an expected value comparison as part of the decision analysis.